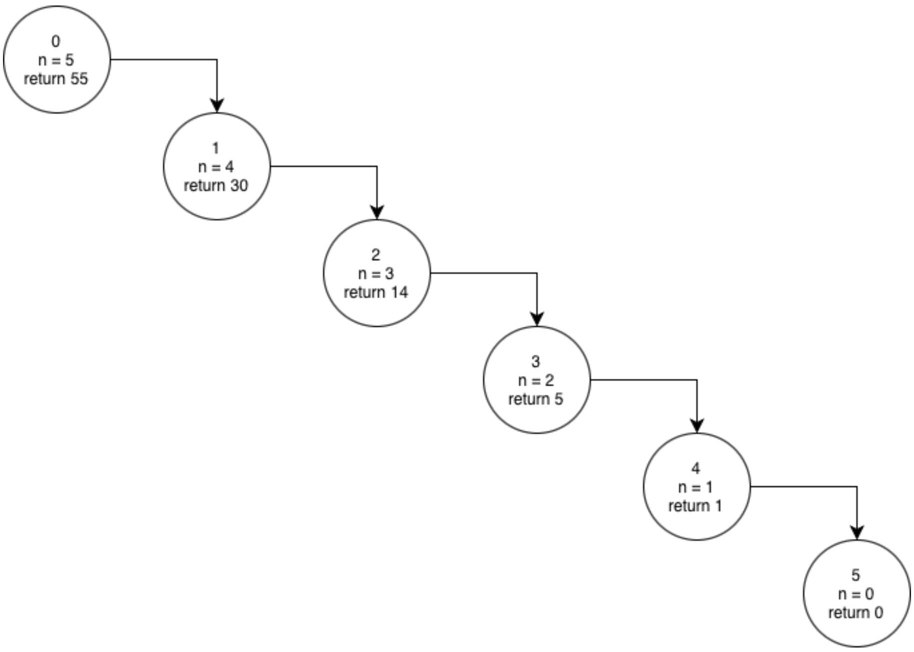
Registration No. 2025/CS/079

Index No. 25000799

(01)

Function call	n	Return value
0	8	384
1	6	48
2	4	8
3	2	2
4	0	1

(02)



(03)

I./II.

Function	P	n	Return
0	1000	5	32000
1	2000	4	32000
2	4000	3	32000
3	8000	2	32000
4	16000	1	32000
5	32000	0	32000

III. 1000

IV. n being equal to zero (n == 0)

V. function calls itself while multiplying the value of P by 2 and decrementing the value of n by 1, until the value of n becomes zero and return the value of P.

VI. function calls itself while dividing the value of P by 2 and decrementing the value of n by 1, until the value of n becomes zero and return the value of P.

VII. value of P

VIII. 32000

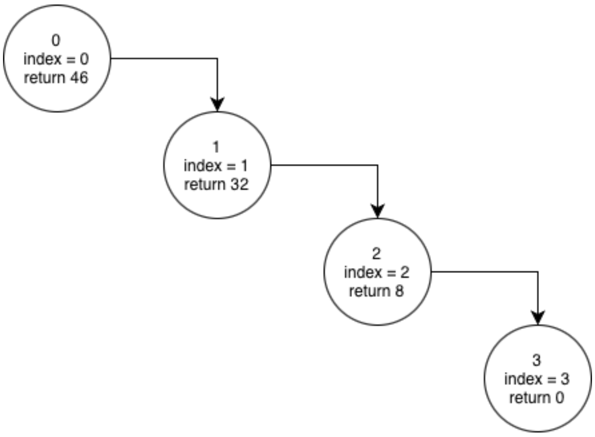
IX.

```
def bacteria_growth(P, n, p):
    if n == 0:
        return P
    else if P <= p:
        return bacteria_growth (p, n – 1, p)
    else if P >= 2000000:
        return bacteria_growth (P/2, n – 1, p)
    else:
        return bacteria_growth (P*2, n – 1, p)
```

Function	P	n	p	Return
0	1000	10	100	1024000
1	2000	9	100	1024000
2	4000	8	100	1024000
3	8000	7	100	1024000
4	16000	6	100	1024000
5	32000	5	100	1024000
6	64000	4	100	1024000
7	128000	3	100	1024000
8	256000	2	100	1024000
9	512000	1	100	1024000
10	1024000	0	100	1024000

(04)

I./II./III.



IV. [7, 12, 4]

V. value of index being equal to length of the array drone_distance

VI. function keeps calling itself while multiplying the current distance by 2 and adding the previously calculated distance, until the index value becomes equal to the length of the array.

VII. 0

VIII. 3 calls

IX. 46 units